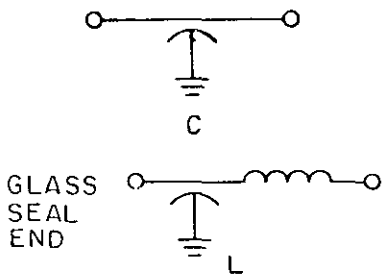
MILITARY SPECIFICATION SHEET

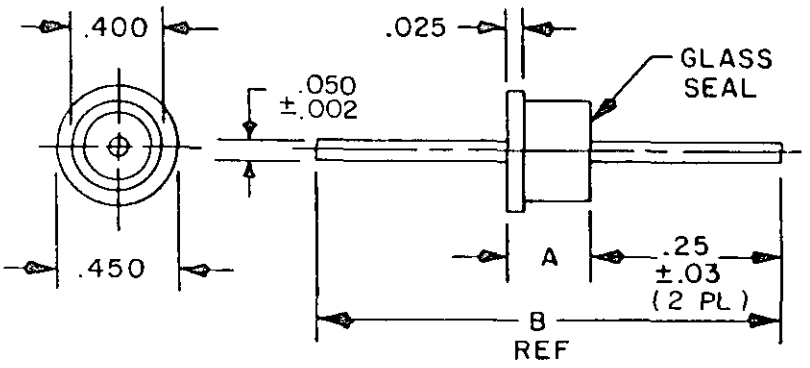
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED ON ONE END ONLY, STYLE FS72

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.

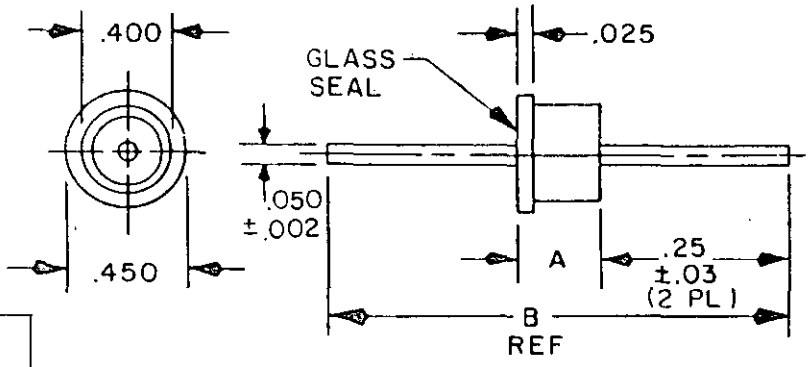


CIRCUIT DIAGRAMS



CONFIGURATION A

Inches	mm
.002	0.05
.005	0.13
.025	0.64
.03	0.8
.050	1.27
.200	5.08
.250	6.35
.300	7.62
.400	10.16
.450	11.43
.700	17.78
.800	20.32



CONFIGURATION B

Circuit diagram	Dimensions	
	A ±.005	B REF
L	.300	.800
C	.200	.700

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005.
4. Circuit diagram is for information only.
5. Filters shall be supplied with 60/40 solder perform.
6. Potting on nonhermetically seal end shall not extend beyond .030 inch (0.76 mm) from the filter body.
7. Filters shall be installed using the recommended installation methods (solder-in style) of MIL-F-28861.

FIGURE 1. Case and hardware dimensions and circuit diagrams.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1. Filters and capacitors shall be hermetically sealed on one end as shown on figure 1 for the respective style.

Weight: .25 gram, maximum for C circuit parts.
.75 gram, maximum for L circuit parts.

Case and lead finish: G only (gold-plated).

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 15 amperes, maximum.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3.0 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm-microfarad or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarad or 10,000 megohms minimum, whichever is less.

Insertion loss: At +25°C: In accordance with table I.
At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: .15 volt dc, maximum, .234 volt ac maximum for dash numbers 007, 008, 015, and 016.

DC resistance: 0.01 ohm, maximum.

Seal: Not applicable.

Temperature rise: +35°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861 except temperature of solder shall be 300°C ±5°C.

Solderability of soldering heat: In accordance with MIL-F-28861 except temperature of solder shall be 300°C ±5°C.

Solderability of mounting termination: In accordance with MIL-F-28861 except temperature of solder shall be 300°C ±5°C.

Product assurance level: Classes B and S.

Part number: The part number shall be as follows:

M28861/13- XXX X X

XXX: Dash no. per figure 1 and table I

X: Case finish

X: Product assurance level per MIL-F-28861

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TABLE I. Electrical characteristics.

Dash number		Circuit	Product assurance level		Rated voltage dc volts		Capacitance (uF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
Configuration			Class					500 kHz	1 MHz	10 MHz	100 MHz	1 GHz	10 GHz
A	B		B	S	DC	AC rms							
001	009	L	X		50	-	1.2	33	38	53	70	70	70
002	010	C	X		50	-	1.2	33	37	52	70	70	70
003	011	L	X	X	70	-	0.70	30	38	42	62	70	70
004	012	C	X	X	70	-	0.70	30	36	41	60	70	70
005	013	L	X	X	100	-	0.50	26	34	44	60	70	70
006	014	C	X	X	100	-	0.50	26	34	42	58	70	70
007	015	L	X	X	300	115	0.05	7	15	34	44	70	70
008	016	C	X	X	300	115	0.05	7	15	34	42	70	70

1/ Insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz. Insertion loss measurements above this frequency range shall be made under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

Marking: Class B capacitors and filters shall not be marked. Class S filters and capacitors shall be marked with a green dot in accordance with color number 14187 of FED-STD-595. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.

Cataloging information: Dash numbers 002, 004, 006, 008, 010, 012, 014, and 016 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 003, 005, 007, 009, 011, 013, and 015 shall be cataloged under FSC 5915 as radio frequency interference filters.

SUPERSESSION: This specification sheet supersedes drawing number 94080, FSCM 14933 when a qualified source is available.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0051-8)

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